STUDYFLOW | SQL QUICK VIVA GUIDE

MySQL 8 database commands, schema and relationships

A compact revision sheet based on the StudyFlow database setup and SQL history.

1. Commands used to set up the database

These are the setup commands visible in your MySQL Workbench history. The password is intentionally represented by a placeholder; never put a real password in a report or screenshot.

```
CREATE DATABASE IF NOT EXISTS studyflow
  CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER IF NOT EXISTS 'studyflow_app'@'localhost'
  IDENTIFIED BY '<your_password>';
GRANT SELECT, INSERT, UPDATE, DELETE, CREATE, ALTER, INDEX, REFERENCES
  ON studyflow.* TO 'studyflow_app'@'localhost';
FLUSH PRIVILEGES;
USE studyflow;
SHOW TABLES;
SELECT user, host FROM mysql.user WHERE user = 'studyflow_app';
```

2. Tables in the StudyFlow schema

Table	Purpose	Table	Purpose
users	Account and password hash	subjects	Student subjects/courses
assignments	Coursework and deadlines	examinations	Exam dates and preparation
study_sessions	Planned and tracked study time	availability_slots	When a student can study
timetable_preferences	Session and daily limits	study_goals	Daily/weekly/monthly goals
daily_goal_progress	Progress recorded by date	notifications	In-app reminders
password_reset_tokens	Single-use reset token hashes		

3. Important database design points

Primary key (PK): uniquely identifies each row; your tables use an auto-incrementing `id`. Foreign key (FK): links related tables.

Relationships: one user has many subjects, assignments, exams, sessions, goals and notifications. A subject can be linked to many assignments, exams and sessions.

ON DELETE CASCADE: dependent rows are removed when their parent is deleted. ON DELETE SET NULL: keeps the child row but clears the optional reference, such as a session when its linked subject is deleted.

Constraints/indexes: UNIQUE prevents duplicate values (email; subject name per user); NOT NULL requires a value; CHECK validates conditions (for example, end_at > start_at); indexes speed up common lookups by user/date.

SQL TOPICS TO EXPLAIN

4. SQL command categories

SQL is used to define database structures, read and modify records, control access and manage transactions.

Category	Meaning	Examples
DDL	Define or change database structure	CREATE, ALTER, DROP, TRUNCATE
DML	Insert, change or remove records	INSERT, UPDATE, DELETE
DQL	Read/query records	SELECT
DCL	Manage database permissions	GRANT, REVOKE
TCL	Manage transactions	START TRANSACTION, COMMIT, ROLLBACK

5. Core concepts in your schema

Topic	StudyFlow example / explanation
Data types	`VARCHAR` for names/email, `TEXT` for notes, `INT` for minutes, `DATETIME` for dates/times, `TIMESTAMP` for created/updated times, `ENUM` for fixed statuses, `BOOLEAN` for true/false.
PK and FK	Primary key identifies a row. Foreign key enforces a valid relationship, e.g. `assignments.user_id` references `users.id`.
UNIQUE	Prevents duplicate email addresses and duplicate subject names for the same user.
NOT NULL / DEFAULT	NOT NULL requires a value. DEFAULT supplies a value when none is provided, e.g. a default status or target minutes.
CHECK	Validates a condition, e.g. exam preparation progress cannot exceed 100; session end must be after start.
Index	Improves search/filter performance; it is not a table or a relationship.
JOIN	Combines rows from related tables using matching keys.

6. Useful SELECT queries for your teacher

```
USE studyflow;
SHOW TABLES;
DESCRIBE assignments;
SHOW CREATE TABLE study_sessions;
SELECT id, name, email, created_at FROM users;
SELECT id, name, code, weekly_hours FROM subjects;
SELECT title, due_at, priority, status FROM assignments
  ORDER BY due_at ASC;
SELECT title, exam_at, preparation_progress FROM examinations
  ORDER BY exam_at ASC;
```

Important: Do not display or share `users.password_hash`. It contains password hashes and is sensitive. Never run DROP or TRUNCATE just to inspect data.

PRACTICAL QUERIES + VIVA

7. CRUD and reporting examples

These are representative SQL examples for the StudyFlow tables. They illustrate the operations used by the application; they are not claimed as literal statements copied from your Workbench command history. Use valid IDs from your own database.

CREATE / INSERT - add a subject

```
INSERT INTO subjects (user_id, name, code, weekly_hours)
VALUES (1, 'Data Structures', 'CSE-211', 4.0);
```

READ / SELECT - list a user's assignments

```
SELECT title, due_at, priority, status
FROM assignments
WHERE user_id = 1
ORDER BY due_at ASC;
```

JOIN - show sessions with their subject name

```
SELECT ss.title, s.name AS subject, ss.start_at,
       ss.duration_minutes, ss.status
FROM study_sessions AS ss
LEFT JOIN subjects AS s ON s.id = ss.subject_id
WHERE ss.user_id = 1
ORDER BY ss.start_at;
```

UPDATE - change an assignment status

```
UPDATE assignments
SET status = 'completed'
WHERE id = 1 AND user_id = 1;
```

DELETE - remove one notification safely

```
DELETE FROM notifications
WHERE id = 1 AND user_id = 1;
```

Aggregate - count assignments by status

```
SELECT status, COUNT(*) AS total
FROM assignments
WHERE user_id = 1
GROUP BY status;
```

8. Transactions and safe database access

Transaction: a group of statements treated as one unit. `COMMIT` saves the work; `ROLLBACK` cancels uncommitted work. Use a transaction when multiple related writes must succeed together.

Parameterized queries: backend code should pass user input as bound parameters (mysql2 uses ? placeholders), not concatenate input into SQL. This helps prevent SQL injection.

Data isolation: queries must filter by the authenticated user ID on the server. Do not trust a user ID sent by the browser.

9. Likely teacher questions

What is MySQL?
A relational database management system (RDBMS) that stores data in related tables.

Primary key vs foreign key?
A primary key uniquely identifies a row; a foreign key links it to a valid row in another table.

DELETE vs DROP?
DELETE removes selected rows; DROP removes the database object itself.

WHERE vs HAVING?
WHERE filters rows before grouping; HAVING filters grouped results.

INNER JOIN vs LEFT JOIN?
INNER JOIN returns matching rows only; LEFT JOIN keeps every left-table row, even without a match.

One-sentence project explanation: StudyFlow uses a MySQL relational database with 11 tables. Users own their academic records; foreign keys connect subjects, assignments, exams and study sessions; constraints protect data validity; indexes support common user/date lookups.